



TECHNICAL DATA SHEET

ProLine PPA-CF Filament

Product Overview

ProLine PPA-CF is a polyphthalamide-based material reinforced with 10% carbon fiber. It combines high thermal stability with low moisture uptake and reduced shrinkage. Parts printed from PPA-CF offer strong tensile and impact performance and can reach a heat deflection temperature of up to 220°C. Thanks to its high-temperature creep resistance, dimensional stability and electrical characteristics remain comparatively consistent across changes in humidity and temperature.

Key Features

- Rigid, dimensionally stable parts
- Strong, durable components
- Performs reliably at elevated temperatures
- Good creep resistance for long-term loads

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm ³	1.21-1.23
MFR (280°C / 5 kg)	ISO 1133	g/10 min	35-40
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.1
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa	108-112
Elongation at break (X-Y)	ISO 527	% MPa	3.5-5 7100-
Flexural modulus (X-Y)	ISO 527	MPa	8000 210-
Flexural strength (X-Y)	ISO 178	MPa	221 70-75
Flexural strength (Z)	ISO 178	MPa	6500-7000
Flexural modulus (X-Y)	ISO 178	MPa	2520-2620
Flexural modulus (Z)	ISO 178	kJ/m ²	9-11
Impact strength (X-Y)	ISO 180	kJ/m ²	2.5-3
Notched impact strength (Z)	ISO 180		
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	220

Continuous Use Temperature	IEC 60216	°C	120
Electrical characteristics			
Surface resistance	IEC 60093	Ω	≤ 10 ³

Recommended Printing Parameters

Parameter	Setting
Print Temperature	290-310°C
Bed Temperature	100-120°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	0-30%
Printing Environmental Temperature	Room temperature to ~60°C

Drying & Handling Notes

- **Pre-dry:** 120°C, minimum 8 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality
- **Annealing:** 120-140°C, 6-12 h to increase part strength

Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

Test Specimens
Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

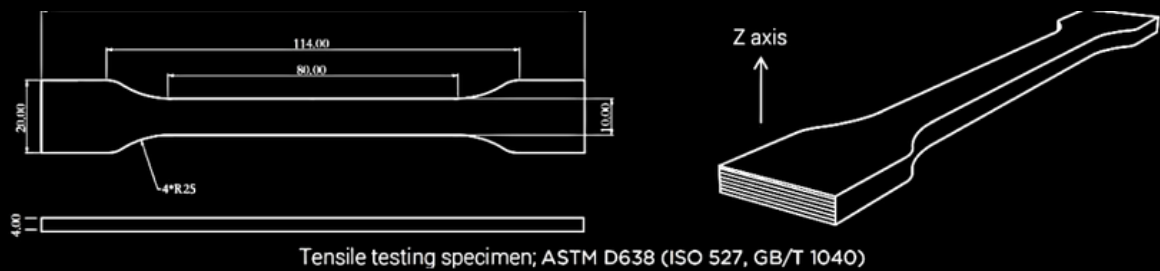


Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)

